

Active inference explained with Prof. Karl Friston

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hello everyone and welcome to ml dawn so today we are having an interview with professor carl frisson uh professor frisson is a theoretical neuroscientist and a world-renowned authority on brain imaging he has invented statistical parametric mapping or spm which is a software package designed for the analysis of brain imaging data sequences and also i think his main contribution uh to theoretical neurobiology is the free energy principle for action and perception that hopefully we'll get the time to also explore professor frisson thank you so much for for being with us with us today my pleasure um okay um so let's just get started the free energy principle right um could you just tell us about what the free energy principle is and how has it enriched our understanding of the brain as a unifying brain theory if you will right so i normally respond to that question by asking whether you want the sort of the high road or the low road approach to um understanding what the free energy principle is about um i'll very briefly try to surmise both so the low road uh which is like a sort of bottom-up way of understanding it which is a kind of way that i think a lot of um neuroscientists and psychologists would understand things or more broadly people in neurobiology would be the denouement at the present time of very old and compelling ideas about the brain as literally a fantastic organ a an organ that constructs hypotheses explanations fantasies that afford the best explanation for all the sensory impressions all the sensory data that we have to assimilate and literally make sense of um so the principles that underwrite that perspective were articulated in various disciplines by people like kant and helmholtz department richard gregory nancy spy synthesis um perception as hypothesis making leverage to greater effect in machine learning with people like uh jeffrey hinton and peter diane and the notion of the helmholtzmann sea machine in turn borrowing technical ideas from the physics and engineering literature such as feynman's free energy principle as a way of writing down a prescription or a normative account of this kind of sense making and you end up with a a formal first principle account of sense making which you can write down as a computer program or you can write down mathematically but essentially casts the brain as in the service of trying to minimize its

variational free energy and you can read variation free energy here essentially as a kind of prediction error technically the free energy gradients are prediction errorism so what's the prediction well it's just the difference between what given your beliefs about the current state of affairs out there beyond the skull-bound brain what would i expect to see what would i predict and then i compare my predictions with what i'm actually sensing and that constitutes a prediction error and then i use the prediction error to drive my beliefs in a way that eliminates that prediction as i got a sufficient account of what's going on and that's often referred to as predictive processing in its inactive form or active inference and when we're drilling down just on the perceptual synthesis that is of you know afforded by this account um the most popular scheme that i would be predictive coding sort of the reciprocal exchange of top-down predictions from the inside out to the parts of the brain dealing with sensory processing that is complemented by a an ascending outside in flow of prediction errors where you need the predictions to form the prediction errors and then the predictions or the expectations they that are generating those predictions um um are informed and updated by the prediction errors technically that's a process called base in belief updating hence the bayesian brain and the helmholtz machine formulations of it so that would be an account of the free energy principle really as a prescription an algorithm a technological account of a normative theory for the brain as an active organ making sense of data with the key twist that of course the brain or you and i are in charge of the data that we're trying to make sense of there's an added inactivist twist here it's not just that we're good bayesian filters or data assimilators we actively have to decide what kind of data we want to base our inferences upon um literally in terms of where i was where i'm going to look next how i visually palpate the world or you know which news channels to listen to or which wikipedia page i'm going to forage next that you know so that that's possibly more um more of a difficult problem the brain has to solve than just simply making sense of the data but both are bound up in this um in this sort of generic account in terms of free energy minimization or the explaining away and accounting for providing the best explanation that eliminates uh united's prediction errors very briefly the high road is the kind of uh approach um that a physicist would uh take and you can derive exactly the same mechanics of belief updating the same sort of [Music] explanation for sentient behavior where behavior is important because that's the active part uh just from an analysis of things that exist in in the sense of physics and the particular physics here is the physics of non uh non-equilibrium steady state system systems that have the the the remarkable property of returning to states that they were once in you know mathematically have attracting sets or characteristic

states of being that you know i am found at this temperature in this kind of location and this sort of environment so all the things that would characterize me represent states that i keep on revisiting and that can be written down mathematically in terms of the dynamical systems theory of attracting sets and you can work out the probability distributions or you can appeal to fireman's work on path integral formulations um to work out well if things exist then what properties what dynamics must they have been must they possess in order to exist and to maintain themselves in within these viable states of being so in the life sciences this would be known as autophagous or self-creation in the computational chemistry of the self-assembly how do molecules assemble themselves and retain their sort of morphology and structural or configurational um organization and it turns out that you end up with exactly the same equations that you're trying to you know everything that we do either on the inside or in terms of acting upon the world is in the service of minimizing this variational proxy for prediction error but also mathematically happens to be quantity that statisticians would try to optimize namely the marginal likelihood or the evidence for models of the world so you get this sort of um technological account um that systems that self-organize look as if they're trying to maximize the evidence for their models of the world sometimes people reflect that as self-evidencing so that would be a philosophical account of the top-down physicists explanation for the way that you and i actively organize our exchanges with the lived world in order to in order to exist to maintain our existence um some generalized homeostasis that can be described mathematically as a minimization of um variation free energy okay uh very very interesting so as far as i understand uh so we are it's about as you said we're trying to sort of do actions that would maximize the probability of our existence of our model that we have from the the the environment that we're living in um so which is i think the same thing that uh that you refer to as marginal likelihood yes the probability of uh observation given the model that i currently have um of the of the world um what i'm trying to understand is trying to marry that idea with the with the other idea that the brain selects which kind of which one of these inputs to consider like how to affect the environment because when we say the probability of my input from my receptors given the model i have from from the environment it is as if i have a control of what i will necessarily do because i mean there are so many receptions coming i might choose to ignore some of them but then it might not be the right choice also when you talk about optimization i realize that now from if i look at this from a machine learning perspective since we're optimizing could we overfit i mean and fail to generalize okay well that's a great questions they're about they're about so first of all just yeah i am sorry i will use

technical um terms with with very little theory of mind because i'm so used to to doing so so you're absolutely right the marginal likelihood is nothing magical it's just the probability that um these sensory data would be solicited experienced um under my model of how those data were generated you're absolutely right the marginal bit comes from um marginalizing out over or averaging over the unknown parameters of a generative model but we don't need to worry about that you know the model evidence is the marginal likelihood and the logarithm of that is is a negative free energy um but your more interesting question i mean if you now um review well you know this this podcast um and just listen to the words you use they're very telling so you talk about selection that's a great word to use you also said i you know so immediately there's a there's a an inactive a generational aspect to your question so a predictive coding scheme that's just assimilating data and making sense of data has no notion of of me has no selfhood and certainly doesn't have any it is not equipped with the ability to choose and to select which data it wants to actually assimilate or to classify or to categorize so we're going beyond machine learning now we are going beyond um simply passively um in a sort of um outside in way making sense of data we are now asking about this really deep question how on earth do we select the data to assimilate the answer is very simple um in the same spirit that um everything on the inside that's doing this assimilation has to minimize the prediction error sometimes cast in terms of surprise in a technical sense so this this is if you're an information theoretician this will be the self-information um if you're doing that then the natural question then arises well look what happens if i'm in control of the data that i can i can now select or turn towards well now i have to choose the actions that will produce data that are have the minimum expected surprise or the minimum expected so if surprise which is another word for a mathematical uh expression for this um bound on surprise or evidence called variation free energy um is the thing in the moment then after i have have acted in the future then what i want to do is to minimize my expected surprise my expected prediction error so what's that well expected surprise is technically entropy and more anthropomorphically uh this would be just uncertainty so what you're doing is if you're subscribing to the free energy principle and you now have to choose the best kind of act or sampling or selection um then you would select those actions that minimize your uncertainty the resolve your uncertainty another way of expressing that mathematically is that they have the greatest information gain that those kinds of data resolve my uncertainty more than those kinds of data so that's a actually a well-known bayesian principle um and it's um often cast in terms of optimal bayesian design and has been known um since the 1950s through the work of people like lindley there are well-defined

objective functions which are simply how much do i shrink my uncertainty and technically measured in terms of a divergence mathematically but you know we just need to know the information gain is just the degree to which i i shrink my uncertainty about states of affairs out there so what that means is that we are if we minimize if we choose those actions or we want to minimize our free energy then we have to choose those actions and minimize the expected free energy which necessarily makes us into curious creatures now you were saying to me i don't know how to do that i don't understand how to that you do you're a scientist even if you don't have a phd every every child and every student and every every adult is a curious scientist they want to know how their world works either it's your scientific field of study or if you're a little baby it's just how your body works you know what kind of thing is mum is mum your same kind of thing as me all of this information um has to be carefully selected in order to resolve uncertainty about this hypothesis on that hypothesis in exactly the same way that you would design a good experiment so the when you design an experiment then you design the experiment to yield data that are going to be maximally informative in relation to your null versus altered hypothesis so it's exactly the same principle governs the selection of natural experiments that we deploy with our eyes looking over there or looking over there it's exactly the same mathematics as the same imperatives and indeed um you actually see the same information theoretic constructs emerging in the visual search literature in the neurosciences and you can construe these as salience maps that you know they score uncertainty reduction the epistemic affordance of making that select selecting those data by by looking over there or by again looking at this wikipedia page or listening to this news channel and not that fake news channel it's a great question because not only does it um does it make the i think the really profound point that that curiosity of all kinds is baked in to the first principal account of self-organization that anything that exists must impart be acting or appear to act look as if it is acting to resolve uncertainty about the exchanges with its economic environment its culture or the visual scene the visual scene at hand furthermore you're talking about action it's you that's acting so it's it there's an implicit sense of of agency that is owned by the artifact in question so we're now talking about something that would be necessary to equip an agent with some a very elemental selfhood the third thing it brings to the table is is really common sensical but it's it's a i think a very interesting observation and that in order to select this action over that action then i have to have a model of the consequences of my action but the consequences live in the future which says that your genetic models now must cover the future which means that you've got something that a lot of things in this universe don't have which is a model of the

future it has temporal depth so you know that you could apply the free energy principle to a thermostat you could apply it to a virus and it would work perfectly well you could simulate thermostats and make a thermostat or simulate a virus or possibly even construct one but these kind the genitive models that you would use that has to be in place in order to work out the surprise due to the predictions generated by that model would not have the temporal depth of the models that you and i have so as soon as you talk about selection you're talking about essentially selecting a plan of action a course of action and that means necessarily rolling out into the future so your genitive models have this temporal depth and furthermore as soon as you entertain the notion that i could do this or that then there is an act of selection in play the thermostat doesn't have that there's only one thing it can do because it acts in the moment but you and i are very have a much more uh a deeper generative model we bring to the table that allows it now to set amongst a series of counterfactual courses of action you can choose what to say next or not to say anything at all and in that choice there is an act of bayesian technically an act of bayesian model selection or policy selection so now you've got thermostats that select provided they have these deep charity models that sort of free you from the moment because they're really about trajectories and paths you know how we navigate the lived world and the about the overfitting thing oh yes could we overfit yes actually why don't we over fit maybe that's the right question that's a good question uh yeah that was the fourth clever part of your question but yeah the reason you don't ever fit in uh and i'll probably um if you indulge me i'll answer this from the point of view of sort of machine learning um or statistics so um the answer here um lies in an understanding of the nature of models and their evidence so the first thing to say is that there is no true model of the world there is no there is a best model in the world and that's simply the model that has the greatest evidence but there's no true model in the world there are you know they're all there are many many ways of explaining um um how some data for us sensory data um was generated and you know providing an account of that all we can do is find the one that has the greatest evidence the greatest marginal likelihood the greatest probability of the data under that model so we can change the model and until the model provides or account gives you the greatest marginal likelihood so what is evidence this is the key point evidence is accuracy minus complexity which means that to find the model with the greatest evidence is to provide an account of these data in the simplest way possible and that simplicity underwrites the ability to generalize in the absence of that complexity part of the model you would certainly overfit so if you did if you were allowed to use as many degrees of freedom um as you wanted with no constraints on

that i know complexity constraints you would certainly overfit these data so that the next day's data would uh not be extendable because you'd ex you'd fit all the little random effects and fluctuations in the in the data at hand so that complexity term is absolutely crucial in ensuring that maximizing model evidence surprise minimizing free energy all of these are statements essentially the same thing goes hand in hand with an automatic penalty on the complexity that precludes overfitting and it's really interesting just to sort of unpack what complexity is here so um on a um on a technical account the complexity is basically the degree to which i change my mind having seen some data so technically it's the difference between some posterior bayesian beliefs some probability just a probabilistic explanation for some causes of some data in machine learning this might be called some the latent states generating unobservable latent states hidden states that are generating observable measurements or observations or sensory sensory input from our perspective so i've got two beliefs before we were talking about this process of dynamically reducing free energy or minimizing prediction error as a process of belief updating that belief updating is literally updating a prior belief before i see data into a posterior belief after i've seen a posteriori after i've seen the data and the degree to which you have to move your belief from the prior to the posterior is a complexity so it's the degrees of freedom that you're using up in explaining the data now if you've got really good prior beliefs you don't have to change your mind very much and therefore you can provide an accurate account of this data with a minimal complexity cost because you've moved your belief updating has been extremely efficient and minimally complex so it is those kinds of high evidence generative models that have this complexity penalty in place that are able to generalize to new data because you've got the right kind of priors so it all goes hand in hand and it's a really important point you know i don't know i i'm wondering whether you knew this already in virtue of the fact you asked the question but you know this this this issue is really um uh you know a difficult issue uh in conventional machine learning because you know if you just look at the way the field has gone in machine learning they've gone to sort of big data you know sort of you know committing to the the ideology of you know big data science with massively over parameterized neural networks so if you read a neural network as a generative model certainly in the context save an auto encoder then what you're starting with is an overly parameterized overly complex genitive model which can beautifully explain any given data set but it's hopeless in generalizing to another context or or some new data simply because the complexity is too high because there are too many parameters in that model that's only say connection weights in a in a current neural network or a convolutional neural network

variation auto encoder what that leads to then is um all sorts of problems with sharp minima that ensue because you have not put in the the complexity term um that you can think of in terms of just occam's principle you know so occurs principle has to get into the objective function so if you've got so the game now it would be to take a neural network in machine learning say in deep learning and try and find out which collection weights or which you know what architectural aspects of your your network you didn't need to provide an accurate account which is actually the opposite direction of travel from most of deep learning at the moment which are aspiring to more parameters with bigger computers there's an interesting little twist there of course that um there are a number of um fundamental principles such as landauer's principle the jarzinski equality which effectively say that information is energy and energy is information and in this present context the degree of belief updating as scored by the complexity cost has a thermodynamic cost which means if you're recognizing things properly and selecting the right things to do properly with the minimum of complexity you're doing it with the greatest efficiency so you can measure the quality of any artifact any neural network or neuronal network simply by the amount of energy it has to expend to exist so again you're the drive towards high performance computing is exactly in the wrong direction what you're looking for is sort of um sentient edge computing like devices based upon a an evidence maximization or free energy minimization approach that should do it um like your brain which would consume the same energy as a light bulb as opposed to a power station so that's a really good question this you know you know that has some really pragmatic engineering um computer engineering and uh sort of you know um energetics uh associated with it and explains to a great extent why you're so much better than any neural network that lives on a a high performance computer at the moment the idea of overfitting and over parameterized models just reminded me of this paper by professor mikhail belkin um so it's it's a it's a paper about this phenomenon that happens in over parameterized models called the double descend as they've called it now their question was why over don't overfit and does why why why they can do why they can generalize so well and it's an it's a very interesting paper it was really shocking to me because we have this famous uh i mean in machine learning we have this famous u-shaped sort of uh you know plot that shows that okay the more you parameterize your model the um well the error in the the error in your in the predictions of your model keeps going down and down and down and after a certain point it fails to generalize so it just keeps going high and higher and higher and higher now what they did was they didn't stop there they kept all adding more parameters they kept adding more and more now what

happened then was after a certain point uh we have a double the a second descend in the generalization ability or in the error let's say the prediction error it again keeps going down and down and down in some cases it goes even below the point where the the minimum of the first descent it could it could even get to better generalization and in some cases it the minimum that it could ever reach would not be as good as the minimum of the first descent so they've called this double descend and i i'm trying to and didn't happen oh the devil doesn't always happened based on their experience they used it with decision trees they used it with neural networks and a couple more models i don't remember exactly what they were but the id the notion of accuracy minus complexity and that being the idea of that being the reason as to why uh we don't overfit i'm trying to uh sort of mix like to have that next to the idea of because their their argument was that when you have too much too many number parameters the model sort of goes towards like you have you have more knobs to control your search in the hypothesis space you you inevitably go to you as if you can you can you can search better and then you find maybe i think they were arguing that you find simpler solutions i think if i'm not mistaken how would you marry that idea with the idea with this accuracy minus complexity being the reason of not overfitting right yeah that's a very question that one okay so um i think if you think of this um the problem in terms of um finding minima in an objective function landscape then that that phenomena of double descent makes entire sense in the context of escaping local minima so let's assume that we've basically got um a difficult problem say a brittle dynamical system or a highly non-linear problem where the objective function let's assume it's either um a free energy objective function or some approximation to it um then it's likely going to have lots of local minima around your one instance and one particularly problematic aspect um you know probably your your your you have in mind is this notion of sharp minimum other minima that have very steep sides that are very difficult to escape from with any sort of stochastic annealing or any sort of um device that would sort of bring you out of that base of attraction of that sharp minimum the first point to make before i pursue that analogy and really basically just reiterate the solution that you you've actually just just described which is absolutely correct i mean i'll i'll do it you know from from a dynamical perspective um the first point to make is that the failure to put the complexity term into the objective functions used by typically by lots of vanilla machine learning not all i mean high-end uh um variation auto encoders use exactly the variation free energy um or which is um in machine learning known as an evidence lower bound or an elbow that is exactly the same um objective function we're talking about but those um those algorithms that don't do that and just

try to maximize the likelihood as opposed to the marginal likelihood so you know just try to maximize the accuracy such as scored by the sum of squared residuals for example so they're forgetting about the complexity and they're just drawing down the accuracy part of it to create these sharp minima the reason that these minima are sharp is that the complexity term creates it into a u-shaped minimum and this is if you like one expression occurs principle that the good the good the good explanations have to allow for some latitude um so the curvature of the minima scores the latitude you have in the sense that you're not committing to a very particular explanation that provides an accurate account of these there are a number ways of understanding this one of them is in fact from the point of view of uh james's maximum entropy principle that under constraints afforded by say a likelihood and surprise on a generative model the best explanation is always the belief that has the maximum entropy which means it has the minimum curvature in some free energy landscape so the free energy functional provides an objective function where all the minima are flat thereby eluding the sharp minimum problems that plague most conventional machine learning so i think that's the first thing to recognize that you will find those free energy minima um more efficiently provided you can escape the local minima simply because the free energy minima are always flat and then they're less likely to trap you um so let's now take what what are what other ways uh could you escape sharp minima um that are local minima that are high in a free energy landscape so just so one sort of picture i use for my students is is very much like a mountain range um that is the genesis of a little stream that turns into a river so the lower you get in the landscape the wider the rivers the shallower the valleys until ultimately you get to an estuary and so exactly the same sort of if you like um landscape exists in a free energy operational free energy concept context so stuff high up uh you know all the sharp minima that have a very undesirable high free energy um are the rust stuff in the higher mountains and your job is to get down uh flow down to the nice shallow um global minima that are much that have this um this sort of maximum entropy occurs principle um shallowness to them with no curvature so how might you want to get out of those high hanging valleys those those sharp minima well you you've said it basically if you imagine from the point of view of dynamical systems theory flows on a um on a manifold or a um in this instance a free energy landscape let's imagine we just got one parameter to play with so and we've found ourselves in a v-shaped minima and we can't get out of it but if i add another dimension and now make my one-dimensional um objective function two-dimensional then i can change that fixed-point attractor that corresponds to the point of the minimum the local minimum in question into a subtle point

into an unstable fixed point and i can escape from it so now i've got a saddle point where there are now roots away from that local minimum because as you keep on adding dimensions i.e adding parameters to your model the opportunities for converting a stable fixed point which is a local minima into an unstable fixed point from which you can escape increases geometrically only supremely so i think that's the explanation in conversation with people in machine learning uh and i enjoyed one such conversation with people um at nyu just a couple of weeks ago i i think that that is is probably the story that you would tell to account for this sort of throwing lots and lots of parameters um at a model just to destroy stable fixed points that constitute a local minima so that you can access the global minima and then once you've done that you then prune away all the parameters that you use as your escape routes to get back to the simple model so what we're talking about i think this double dissenting is a really lovely notion and it does remind me a little bit of how the brain works in the sense that you know it starts off with an overly parametrized um genitive model so babies and children actually have more connections they have more w's um more external connections white matter tracts than uh than adolescent something you and i do so you get this progressive pruning as you grow older so you start off with this overly parameterized overly complex virgin tablaraza where you can escape all the local minima and then as time goes on you then do your pruning and reduce the number of parameters or collection uh connection weights in your in your model to to simplify to to uh in accord with james's max venture principle outcomes principle uh from the point of view the free energy minimization it's the complexity part of the minimization so you now get a smaller parameter model but you've only done the second wave of descent if you like after you've escaped all the local minima by having an overly parameterized model so i'm sure that that's you know this must be a universal principle that i suspect is so universal it has been discovered by evolution uh you know millions of years ago yeah yeah that was that was amazing wow that was a totally different perspective that the the i mean the reduction in the number of parameters as you grow old i i i never thought about that first of all i didn't know that but it was it was really amazing yeah well it has its penalties because i am much older than you and therefore i am much sparser and wiser than you but i could not adapt to a new world so i can't go bungee jumping or go to disco oh my god wow wow wow mind boggling i i don't know what to say that was that was really enlightening actually um so it seems to me that uh there are two methods of learning in general when it comes to biological systems so i did a little bit of research on this i don't know whether i'm right or or not but i could narrow this down to two general methods one is perceptual inference where and the

other one is through action inference now the perceptual inference i i suppose is when you generate a model from the world and you keep updating that the parameters whereas the action one is you actually choose actions um so i know we've discussed this a little bit at the beginning of the interview but could you explain these two methods in more detail and and compare them as to it in what situation am i learning through perceptual inference and in what situation do i choose to learn through action inference right yeah it does i think speak to this issue of sort of um machines with agency that we were talking about before artifacts that have sentient behavior so the sentient part would be the sexual inference and the behavior would be the active inference and i think that they you know once you think about sort of artifacts that do actively exchange with the world i think they both have to go hand in hand so um i you know i think it's although another excellent question i will qualify my answer after i've given you the vanilla answer which is you know these are this is just a joint optimization process over two kinds of unknowns two kinds of latent states out there that generate data one kind of latent state is the latent states over which i have no control so this would be the standard data simulation classification problem recognition problem i am given some data and i have to classify learn its causal structure infer the particular state generating latent state generating these data in this particular context so that would be perceptual synthesis perceptual inference and learning the parameters of that generative model i would call perceptual learning so in my world i distinguish between fast processes that infer time dependent states and slow processes that infer the parameters that underlie the contingencies and the laws um we would be learning in the machine learning center so that the w is in a neural network but the states of the network at any one point in time i would interpret as imprints um you know this is the activity of these nodes and therefore i'm seeing this kind of face so that's one set of unknowns but there's another hugely important set of unknowns latent states out there those which i can control those which i have some agency over so if i make this movement or i secrete this or i switch on this knob then the states out there will also change if that's the case then you're now conditioning a certain set of hidden states on latent states which can be thought of as plans or control variables so the you know sort of uh standard machine learning treatment so these control variables or uh say in cultural theory now play a really important role because the transitions the dynamics amongst the latent states um that are controllable are now conditioned upon this u but this u is another random variable this means i have to infer two sets of random variables i have to infer the latent states over which i have no control and i also have to infer control variables if you like the plans um upon which the variables the states that i do have

control over are conditioned so there's two kinds of inference here which you can think of in terms of the distinction between someone just applying a kalman filter to estimate the states of some plant versus um estimating the best thing to do if you you were doing applying control theory and you know when you're controlling a particular plant say in engineering but because we're now committed to understanding everything in terms of optimizing a free energy functional of beliefs or probability distributions that now becomes that that planning or control theoretical part of the problem the active influence part of the problem now becomes planning as inference you know in a sense you can read the two perceptual and active sides of imprints as really the same process working hand in hand in parallel but acknowledging there are two different kinds of variables that you you have to infer you have to work out basically what's the state of affairs out there and what am i doing about it and then of course you can see now that the active infant side the um the planning as inference now has to consider a number of plausible plans and then we get back to selecting the um the most likely plan which minimizes the expected free energy or results which is uncertainty or avoid surprising outcomes like you know um your your machine crashing if you're in an autonomous driving uh situation so you can write in uh all your prior preferences into that um expected free energy in the future conditioned upon the plan i would i would say that that term uh that sort of you know there is no choice to do active inference means that you have to infer both the latent states of the world and the way that you're intervening and uh on those states through controlling state transitions that just means planning as imprints that's part and parcel of active influence and just being and being um existing in a world that you're continually sampling and it's continually generating data for you to make sense of there's you know there's a circular causality there the the the sort of the twist here um the qualification is that i think you're absolutely right that in fact we do switch between that active and perceptual inference in reality so if i was building an artifact i probably wouldn't worry about what i'm about to say but if i want to describe you and me i would describe you and me as basically machines that are built to um in a saltatory way in an intermittent way alternatively plan an action and execute it and then gather some data and then make perceptual inferences or you know do the perceptual learning side of things so i mean this in the sense that um the way that we the temporal scheduling and the alternation between the action and the the perception of the active inference and the perceptual inference the way that it seems to be scheduled and organized in biological artifacts like like ourselves is that every 250 milliseconds about four times a second we act within those actions we do our perceptual synthesis so this you'll see this yo for

example every 250 milliseconds about four times a second you will move your eyes with cyclic eye movements so you're you're getting little snapshots every 250 milliseconds at four hertz um and during the acquisition during the action you actually switch off your sensory channels it's called saccadic suppression if you're an engineer you're basically turning the kalman gain down to zero when you're doing the action bit so you don't see the optical flow induced by your eye movements you just don't see that what you see is the thing that you've actually fixated on and you you uh you use that to construct a scene you assimilate you build you build up your this illusion that the you know there's a visual c down there but in fact in reality that in little snapshots from here and here at about four hertz the way that you speak if you think about or just listen to me and the way that you you hear things um well that's not quite the truth certainly the way that you um you fulfill your proprioceptive predictions during speech all my phonemes are roughly 200 milliseconds long so i'm just producing a second you know a discrete sequence of little packages of phonemes every 250 milliseconds if i was a mouse i would sample sample my world by whisking my whiskers and the frequency of that is 250 milliseconds or four hertz and if you now look at the belief updating on the inside which would correspond to the perceptual inference making sense of the data sampled by my whisking around my burrow and you see again this profound theta rhythm that dominates the active exploration of the environment that within it is nested all the fast perceptual belief updating usually at what's called a gamma frequency about 40 hertz so you've got this lovely separation of time scales where we make a move on the world at your four times a second we get some data and then we quickly process what that means when we've reached a conclusion then we go off and get the next bit of data and then we process this so for artifacts like you and me um there is actually i think a temporal separation of the active part and the the perceptual part part of inference so i'm waiting to see until they build that into robots because i think you can very realistic um behaviors and the way that people respond to things and you know orientate to things or understand things through this sort of discretization or quantity quantization of discrete packets of information that live within say you know a few hundred milliseconds and you know and that dictates the type of scheduling of at least biological uh belief updating unbelievable that was amazing so it it feels like there is no parallelization there is actually very fast switching between the two like a joint optimization yes just require you know if i was building a computer i wouldn't do that it's just that obviously there's a better way of doing it an evolution has found that this sort of uh this fast switching is the best way to do it in the to explain the kinds of worlds in which we live in we are where biological motion has

certain time constants were you know meteorological events or things around us changing a certain time scale it's clearly the most efficient way uh you know according to evolution anyway oh basically oh yeah absolutely wow that was amazing so uh so we we keep we kept talking about active inference and of all the actions that i could choose i'm choosing the one that on average would maximize or say minimize surprise in future or minimize free energy i'm trying to think as to so first of all it it requires i suppose i don't know whether the word is my wording is right or wrong but some sort of a knowledge about the future as you said it requires look into the future um i can't help but notice that there is a similarity between active inference and reinforcement learning however in reinforcement learning what we do is we have the ultimate we like we have an end goal and end state and we just let the machine do whatever it's it wants to do round like exploration exploitation whatever it wants to do many many many times so eventually it has a way to back propagate the expected reward to the current state a human being that say it's experiencing something for the first time and wants to choose the action i i don't it doesn't have that that that option first of all how would you differentiate between active inference and reinforcement learning and second of all let's let's just discuss as i forgot the the second question uh let's just discuss the first the difference between active inference and uh reinforcement learning yeah i relieve because that's a very big question but feel free to prepare the next one while we're well we're dealing with okay perfect so um there are a number of ways of answering that question and the simplest way of course this is to look at what is on the tin when you pick up active inference versus reinforcement learning one's about inference and one is about learning active inference has an influence and learning which has the important consequence that you have um beliefs not just about the um states of the world but also beliefs about the parameters of your genetic model so it comes equipped with beliefs about the w's in say a neural network reinforcement learning is just about learning the parameters you could look at reinforcement learning um as a mechanism to enable machines to learn to infer and in part i think that's how people often um understand inference in the context of machine learning and strictly speaking the optimization process is applied to the weights the parameters of a generative model the contingencies so there's um there's one fundamental difference between reinforcement learning and active inference in the context of i don't think that's quite the spirit of your question though so i think another way of looking at the difference reinforcement learning is just in terms of the teleology and asking what is it that you want to optimize so reinforcement learning you assume the existence of a reward function or loss function a cost function a value function and then you optimize your

neural network or your artifact um to um respond to any given um either explicit state or inferred state with a particular action the and that action is chosen by the nationality principle to maximize the expected reward so that presupposes existence of a value function um and if that exists then by the optimality principle you can then events a an optimal state action policy is that um is it can that be um if you like framed in terms of active imprints the answer is yes but as a very very special case so what is that special case well it's a special case where you can reduce the function that is implicit in active inference which is a functional of belief so remember your planning as inference belief updating in terms of perceptual inference and and data assimilation evidence accumulation and all that good stuff kalman filtering and the action selection via planning as inference this is inference so the objective function is a functional function of a function or a functional and that function is a belief so the objective function in active inference is uh belief based so under what conditions could you actually reduce an active inference formulation of a problem to a bell to the kind of problem the optimality principle would apply was when you shrink your belief down to have zero uncertainty so if you take uncertainty off the table then you can get back to a effectively a situations in which you can apply the bell but optimality principle interestingly just for your entertainment the way that you do that technically is just by describing your reward as a log prior some outcome that you prefer so we often call it a prior preference so we say that every outcome um is equipped with a reward or a loss function so instead of having two separate channels sensory information and reinforcement information we simplify the problem and say no every output every observation or sensory input comes labeled with a rewarding aspect so your reward function now becomes very high dimensional and continuous it's just basically the long probability of a priori me something that outcome and once you write that down and you take away uncertainty um then you can apply the beveled optionality principle so you can look at reinforcement learning as a special case of active inference when there's no uncertainty the problem with doing that though is if there's no uncertainty that means that the curiosity part of the objective the expected free is zero which means you've lost the opportunity now to simulate artificial curiosity and exactly the kind of sentient behavior that we were talking about half an hour ago because there is no information to be gained there is no information seeking at hand because you know everything why do you know everything well because my in my model of the world in my reinforcement learning model of the world there is no uncertainty i know everything i can see all the hidden states or at least i can infer them um the the the what you lose by converting a an objective function which is a function of beliefs

which is you know the variational free energy or the marginal likelihood or the model evidence all of these are usually logarithms of or kl divergences or free energy functionals of probability distributions beliefs if you now try to you know coerce that into a function of states as opposed to beliefs about states you take away all the epistemics you preclude any proper discussion or treatment or account of artificial curiosity in your robotics this would be intrinsic motivation so um you know the the the intrinsic value the value of information the epistemic value of making that move what would happen if i did that has its own if you like epistemic reward but that part of the reward function also more explicitly not part of the expected free energy uh or the expected evidence or the expected uh surprise and that part has um has been removed when you dispense with or you you try to cast the um the problem um in terms of the bellman optimality principle just a little technical interesting technical twist here when you um when you consider the full problem where you're dealing with um functionals or probability distributions for us bayesian beliefs um you are technically dealing with um usually path integrals because you as we've spoken about before you know our paths go into the future and so the plan has has a trajectory or a path into the future so now you've got a path integral of a functional of a log basically a long probability which is an energy so you've got a path interval of an energy which is called an action and you're trying to minimize expected free energy which is this this action so all you're saying is that i can explain sentient behavior with hamilton's principle of least action so now you're replacing the bellman optimality principle which applies to value functions of with hamilton's principle of least action which um deals with if you like value functionals or belief states or beliefs about about states so that's for me how i would see that there's a fundamental difference active inference but it's not a difference that should be overly celebrated in the sense you can get from one to the other you can repair that dialectic if you like as long as you're prepared to take all the reduceable uncertainty out of the game or off the off the table does that is that a kind of yeah yeah yeah um i'm trying to understand one bit of this um so when you say that in reinforcement learning we actually know there's certainty in reinforcement learning sometimes we're not aware of the world i mean like many things could happen that could like put obstacles on on the path of our agent that could happen but then the agent somehow learns to you know get out of them um i i don't get the idea of not having uncertainty and reinforcement i didn't get that part to say that there is no uncertainty in reinforcement learning would be disingenuous and i i think profit possibly for me most easily understood in terms of reinforcement learning um being one application of bayes optimality in the context of optimal basic decision making with a loss

function so it's really it was really the nature of the loss function i was talking about what i was trying to get across was that um bayesian decision theoretic approaches don't have as part of the loss function any reduction of uncertainty so you have to build that in by hand so that novelty bonuses or exploration bonuses and things uh so you have to apply all sorts of widgets or little tricks to to make your own your loss function even treated under uncertainty you know in a basic decision theoretic context or say a partially observed market decision process you have to actually write in what you get for free if you start with an expected free energy or unexpected um marginal likelihood um so it's it's a kind of uncertainty that matters in terms of what i believe i was talking about so it's it's a difference between um uh you know a neural network that's trying to be on target uh for as much time as it can in a noisy uncertain situation for example and a neural network that knows it doesn't know certain things that would enable it to find out what what what particular times it should be from the target or from what particular direction because there are certain unknowns out there that it can actually resolve with particular epistemic moves that have this intrinsic value of this intrinsic motivation um that give it this curious aspect so it contextualizes the optimal bayesian decision theory exactly by adding in the optimal basic design principle we're talking about before which is this which is this sort of uh information seeking aspect so the the full picture from my perspective of the perspective of the free energy principle is that that you would you can decompose the objective function expected free energy into two parts literally linearly um into into you know an epistemic part and as a reward or pragmatic part and that's the epistemic part that disappears when you go straight for optimizing the reward function or the uh or the your prime beliefs there's an interesting intermediate case um which you can get out just by rearranging certain terms in the um in the excited free energy functional which is um kl control so this would be like risk sensitive control so i'm not sure whether you put this under reinforcement learning or not um so i'd have to ask you but so from my point of view this is a an interesting middle ground where you've now included um uncertainty about the consequences of my into the game so instead of now just trying to maximize the expected utility of the expected loss function on the expected reward under some belief about the consequences of action i'm now um going to minimize the kl divergence or the difference between what i think will happen the outcomes that i under my posterior predicted density technically but more intuitively what i think the outcomes will be if i take this plan or this course of action and then i have this set of preferences you know that encode my preferred outcomes and then i just choose the action that minimizes the kl divergence between the two so this has uncertainty um in it um that is

not quite the full info uncertainty reduction but it certainly accommodates a degree of uncertainty um when it comes to real world applications uh again um i'm thinking about sort of engineering applications in control theory where this will be known as kl cultural and economics it's called risk sensitive control so that would be like a finessed reinforcement learning value maximizing scheme that has gotten uncertainty banked into it and you know that's an interesting um uh halfway house between the sort of vanilla expected utility which i'm reading as a reinforcement learning process possibly unfairly um and the full um the full size of active inference um expected free energy it's a really great game if you have time in terms of just writing down the you know the probability distributions and then and just rearranging them to different kinds of divergences and just trying to read them as if you're an economist or you're a an engineer control theoretician or or you're a behavioral psychologist reinforcement learning or your neuro roboticist developmental neuro roboticist trying to understand intrinsic motivation but by switching them around you can get all sorts of interesting different perspectives on this overall um you know expected surprise minimization uh like objective function i see oh that was really amazing so we talked about perceptual inference and active inference now both of those ideas highly rely on the environment one is modeling the environment and the other one chooses actions that would you know maximize the the probability of my existence now both of them rely on the environment but what about when we come up with some abstract knowledge that never existed before like the first time that someone thought about the wright brothers thought about an airplane or the first time some someone thought about a mermaid um so anytime that something if that's even possible i don't know the first time that something jumps into one's mind that never has existed in the and they they try to make it into like create it into reality can we explain that part the knowledge generation with with these two ideas uh active inference or perceptual inference yeah again another very challenging question um i i would contend that you can so i think you're touching now upon the links between the mechanisms of um um the underwrite model optimization and to say curiosity but more i think the notions of creativity and exploring different model spaces so we're now moving in fact actually completely away from active inference and perceptual inference in terms of um optimizing both the states and parameters of a generative model as it is exposed to data and now turning to a third level of optimization which would be the optimization of the model in and of itself in various fields this will be known as structural learning for example in medical and uh constructivism the problem of not they're saying forget about let's assume we've got the perfect scheme if you give me a neural

network or a generative model and a particular environment that can generate data i can do all the good inference uh or learning to infer um actively and all the canada's inference and so we've got this little agent now um coupled exchanging with their or her environment um in in in a base optional way now we move to a next level hierarchy speaking of an optimization or free energy minimizing process which is now not applied to the parameters of the model but to the model itself how many hidden layers how many connections do you use rectified linear or sigmoid trans you know how do you carve things up in terms of not just hierarchical levels but in terms of know factorizing or modulating modules or clinician dependencies at any level all sorts of fundamental structural issues that you would practically have to deal with and contend with when writing down the architecture of your neural network that you know will read as as the structural form of your genetic model so this structure learning problem and from the point of view of a statistician is a bayesian model selection problem and it's basically integrating the marginal likelihood over all the kinds of data that you could see or all the kinds of data that you have seen and then you choose the model that best accounts for that has the greatest evidence so now this is a categorical active selection of selecting the best model and i use those words because this is mathematically i think an apt description of natural selection so natural selection and evolution is just nature's way of doing bayesian model selections is selecting the hypothesis namely the phenotype um where evolution thinks that this particular hypothesis is a good fit to me the clinician that i am supplying um and therefore the econis then selects the phenotype through this process of the structural form of the epigenetically specified for example um structure of the gelatin um and as soon as you think of it like that then then um there's a big question and the big question is well how do you explore the model space because we've just said when you're doing this bayesian model selection after you've acquired the data so now what kinds of model spaces would you would you explore is it necessary to explore there is an argument which we actually touched upon earlier on that we can actually start off with a completely over parameterized model and prune it so this is a kind of basic model selection technically called basic model reduction where you just literally if i remove this parameter of this connection from this model or this hidden layer from this model would i have um would i increase its the pathogram of its uh of its evidence um in other words um would i be would i be minimizing complexity without sacrificing too much accuracy with the kinds of data that this model is fitted to explain so you know you could argue that both the model expansion typically dealt in amazing context through non-parametric bays so by having expandable models and priors of the way that you would add in different parts or chunks so for

example in a hierarchical dirichlet process you might have some sort of stick breaking process on top of that which says you know can i am i is it justified to bringing the new hidden later variable or you know sort of uh in my uh dirichlet distribution uh and you'd ask that question by basically um evaluating the evidence for that model or the variational free energy um with or without that extra bit and if it improves by adding complexity which more than pays its way in terms of the accuracy then you keep it if not you wouldn't but you you know what the what the stick breaking process brings to the table is is a principled way of expanding or growing the model i repeat in the context of non-parametric basing approaches but i think you can also you can also sorry i think that both model expansion and i'm reading that at the moment in terms of um non-parametric base um or from the point of view of natural selection um having recombinations of certain sort of codes um or parameterizations of of um you know of of genitive models that you can think of in terms of split emerge like operations that you get in genetic recombination ways of exploring a model space in a principled structured way or and the basal model reduction i think both can be seen um as the processes that give rise to creativity and insight and aha moments so um i'm i'm saying that are using some of the notion of aha moments and creativity um in um deliberately just to make the point that some creative acts are actually acts of reduction it's not so much the fact that no one's ever seen this before um sorry no one has ever represented or um had this um this um object in mind before it's just that no one's seen that connection before and it's actually a simpler way of viewing things so if you just think about the most creative ideas and signs they are not actually more complicated de novo constructs they are simplifications that make sense of lots of other things in fact they are instances of the free energy principle itself providing a simple account of uh but accurate accounting of everything so it's really the the simplification and the reduction the increase in the evidence when you suddenly see the two things that you separately represented in say two parts on your network can actually be represented by the same thing and you do that by removing it to the model so actually many apps of creation uh certainly in sort of minimalist art forms um are actually getting to the bare essentials of what's going on again in um you know in compliance with occam's principle and you know um you know um advertising the complexity cost of explanations from for the lived world that that seems to me you know an important bucket of creative acts is actually just seeing the structure underneath it's simply the simplicity explaining everything with just one canonical platonic like um dynamic or construct or narrative or or sort of factorization however you want you want to um however you want to um articulate it and in that sense the first person to have that simple insight that oh it's just one

of those this is just one of those the first person is really the genius and the creative person but they haven't created anything new they've just seen something simpler uh i see um and in a sense you know that's i think that's the scientific process that's exactly why we're having this conversation that's what you do with your world so what i do with my life and i guarantee what you do with your life it's this journey of finding the simplest explanation for everything that we that we experience uh and you know leaving it as a legacy you know speaking to the uncultured aspects of uh or um that one could uh appeal to to answer your question uh for the next the you know my children or my students or whatever so you know that's i think the most that's one way of looking at creativity the an interesting point that ensues from that is that um you know it is perfectly possible to be the first art sentient artifact to see that simplicity um that is actually in the real world so it's not the question really of making new things that aren't out there they've probably already been out there it's just you're the first person to have uh to be able to infer them in that simple kind kind of way the other point to make here of course is well is it really out there well probably not this is your simplest explanation it comes back there is no true model it says this is the best explanation for what's going on so you may be the first person to find this simple explanation um for what's going on so i would put most of um most of art and creativity in that in that bracket but then you compress me on music for example and and you know why is why is music so attractive and well that would be another question maybe okay yeah questions yeah perfect um that was amazing actually professor i was watching one of your talks actually it was 3 a.m as i was watching that and you said something that i mean my mom was sitting next to me and you said something that it was so interesting that i i it was like i got a nervous tick or something then my mom asked me what happened i said he said something amazing and i just want to share that with you and just maybe we could dig a little bit deeper in this um you said something along the line if i'm if i'm putting this correctly the way our biological neural network forms itself tells us something about the environment that we're living in in other words show me your nervous system then i'll tell you about the environment you're living in and it's fascinating to me because maybe um a couple months months earlier i was discussing this this gap between artificial neural networks and biological neural networks and i was asking my friends that these distances like why these axons are long or why not or these dandruff we never create those structural uh foundations into our artificial neural network it seems like very simplified version of those what do the what do those features entail really and when i heard the piece that you just said in in that in that talk it was just amazing now impres and you mentioned the idea of action in the

distance and how light you know reflects into our eyes at that how that's how we have long neural you know connections in our like reflection of action the distance now in practice if we actually see like actually in in a literal meaning if we actually see a ner the nervous system of an of a biological being how much can we really say about the environment like is it really possible to to say okay probably it has these rules it has gravity it has whatever is it really possible yes that was a lovely story more interesting than my answer yes yes i think it is you know you've just convinced your listeners it is possible um have you given some of my favorite examples but i'll give you a couple more just to reinforce the point you're making uh i think the first thing there's a couple of things before i um get into this uh you know you were noting that the architectures that um have emerged in neural network theory and machine learning and naturally deep learning have a very simple architecture that is if you like that is inherited from neural networks and that's a good thing so remember simplicity is good and inheriting from neural networks if you want your machines to deal with the kinds of things that human beings deal with those are two good things they may be a bit too simple but i think there are certain architectural features that you can actually spot which actually would allow you to apply your law that you've just suggested give me a neural network and i'll give you the world that that neural network is fit to explain or act for um that law is um another law um from ross ashby we mentioned the law of requisite variety before the context of natural selection um but he um with colleagues also formulated the good regulated theorem so i'm not sure whether whether you're probably too young you're far too young to remember this this is this is um was taught as one of the fundamentals of the inception of cybernetics you know in the early 19th 20th century so the law the good regulator theorem which you can you can wiki there's a wiki little wiki page or basically what it's saying is that if any artifact and from from ashby's perspective this would be something called a homeostat which has all this sort of you know regulatory uh aspects to it that we talked about it if it regulates its environment basically it does the right kind of active controls the latent states out there the right kind of way to survive then it must be a good model of that environment so he was uh and he purportedly or thought that he could prove that and i thought you know i i've read the paper way where he proves it it's not an easy read but if you have a 3am in the morning with nothing else to do you should try and read it it's original fruits let's see see if you're convinced um but the key point is that this is a very old idea um and probably you know a very true idea that anything that um exists um in some kind of generalized synchrony with its economics environment its heat bath its world it's uh um climate uh must effectively be a um be a good model of that climate which

simply means that the architecture the structure that we're referring to in terms of structure learning must in its function in its structural form recapitulate the causal structure of what's out there so the structure of the generative model um will tell you an enormous amount about the you're the kind of world that is generating the sensations that are being used to do the planning as inference to control that particular world so lots of great examples you've given one well i think one of the most interesting ones which is why is the brain why does it have an architecture that involves long thin communications at a distance i mean the liver doesn't and the liver does a wonderful job of doing its job and presumably complies with the free energy principle and doing its own kind of active imprints in a world of chemicals and metabolites you you you've given you've given the viewers the answer it's just that that must imply a certain kind of conditional dependency that has this long-range action at a distance and of course we unlike um thermostats and worms and viruses really have to live in a world where there is actually a distance in the sense that i can hear you from dublin i can see you from across the room now viruses worms and thermostats can't do that they are they their world is is you know is much more immediate and just relies upon the conditional dependencies that are related to the juxtaposition in symmetric space but we live in a much more complicated world that has this action at a distance that that actually is necessary to explain the kind of sensations that we that we have to explain particularly sort of vision and and audition conveyed by waves of one sort or or another or another sort um so i think that's a wonderful example but you can go even further and sort of course the grain you know you just look at the brain and it's and it has two hemispheres that tells me immediately that the embodied world of this particular brain has a bipedal symmetry yeah so you know if the world includes the body then i can tell you almost immediately that the kind of creature you dissected this brain from um the part of the world which is most important to that to that particular brain or neural network maybe in the body that embodies it's it can move has a bilateral symmetry will probably have two arms and two legs and all that good stuff if on the other hand he gave me the brain of an octopus i'll be able to tell you immediately because they have a brain for each arm it has an eightfold symmetry so i can tell you a lot about the the world and in you know in this instance the you know the bodily world uh from just the gross morphometry but you can go much much further uh i could look at your brain and i will find two streams that emanate from the visual parts of the brain and i will know that they're visual parts because your eyes are my central nervous system and i know that they're they're photoreceptors so i will know that they're somehow um trying to explain the causes of sensations um um in a visual scene and i will find two streams

called the dorsal and the ventral stream that sort of run eventually in this direction and also also in this direction actually uh towards the hippocampus more eventually here and that tells me immediately that there's been a factorization there's been some sort of a fundamental carving nature at its joints in terms of the causal architecture of this creature's world and in particular what i will be able to discern is you must believe that certainly that there are the objects have at least two attributes um in order to um produce their impressions of the nervous system in the words of the helmholtz and of course those will be what and where and i'll be able to tell you that you the typically the kind of visual objects that you're trying to will have these two attributes that are conditionally independent because of the physical separation and the um the sparse connectivity between these two streams um so what we're saying just in common sense terms basically means for the typical visual objects that we usually look at knowing what something is doesn't tell you where it is and and me telling you where something is doesn't tell you what it is so it is the most efficient simplest minimum complexity representation just to represent the wholeness and awareness and then you put them together to explain this thing in this in this location another nice example of minimizing the complexity by using the minimum degrees of freedom when carving up the world uh you know in terms of being a good model of that world but you see exactly the same kind of um if you like factorization and effectively weight sharing in convolutional neural networks which brings me back to you know don't don't be dismissive of your cnn your favorite cnn it's got some beautiful factorization uh in it so the very fact that you've got this weight sharing and this sort of way of carving nature at its joints um in the context of what you see emerging convolution your networks is that you are now saying that there is some causal structure out there that has local continuity or contiguity aspects that i'm going to leverage to provide a really simple explanation for these kinds of data so as soon as you tell me i'm using a cnn i know exactly what kind of data you're using you're using stuff that has certain regularities in the metric space spatially extensive uh so image-like data if you give me another kind of neural network that looks like a transformer network i know you're not doing that i know that now you're trying to deal with things mapping one set of categorical uh objects onto another set of categorical objects so just by looking at the structural form of even i think um sort of canonical architectures in deep learning um you can tell a lot about the causal dependencies the independent the conditional independences that generated the kind of data that you're trying to classify or um auto encode i see professor this has been amazing so the final question the last question that is going to be a very non-technical i i hope like like a simple and nice

question for to end this talk a light question that i'm sure many people watching uh would be interested to know your opinion will we ever fully understand how the brain works how far have we come at the moment a simple or is it a [Laughter] long a long difficult one i i appreciate your attempt to ask a light a light fun and i think we've come an enormous way and there is there is an enormous way to go and i think there are some principled issues about the understandability of ourselves that actually get into some deep questions about consciousness and metacognition and the kind of creatures we are and of course the kind of curious creatures we are and have to be in order to exist um so i'm thinking here of um things like david chalmers's meta problem or met a heart problem so now in philosophy the question has moved from away from what is it like to be or to see red really why do we spend all our time puzzling about the fact that we can see red or experience things and of course that speaks to this hierarchical or genitive model where parts of this genetic model now have a construct of meanness and meanness in particular experiential contexts um hence the metacognitive aspects when you think about very deep character models so i'm sorry what does what do you mean by meanness oh so you have something which you could argue um many um lower forms of life possibly not dogs and cats but sort of possibly birds don't have you have a part of your genitive model is a hypothesis or a fantasy that you are a person and that you are in charge of orchestrating your perceptual influence in your deep genitive models and moving and secreting now you don't need to have that hypothesis or explanation in order to function quite viably it's quite a sophisticated being like a bird for example but you have that which suddenly creates the opportunity to think about counterfactual ways of me being so when i said mean us i just meant me in inverted commas less yeah oh beingness sorry okay okay i i okay okay yeah yeah yeah yeah yeah sorry sorry yeah yeah just your question will we ever be able to understand ourselves i think um it's a lovely question in the sense that the answer to that may um hold the secret um uh to um awareness and particular self-awareness mean awareness selfhood and what particular kinds of generative models you would have to have in order to even ask the question um am i yourself and then the philosopher's question can i understand myself these are really really very very literally deep problems that rest about deep hierarchical gender models i see perfect uh thank you so much professor fristen it was indeed an honor to have this uh talk with you i really enjoyed it i hope that i didn't try uh tire you out too much it i know it it took a long time he did and i'm exhausted but thrilled to have spoken to you i really enjoyed talking to you thank you so much thank you so much professor and i'm truly truly sorry for taking too much because taking too much of your time because it's like

one interesting argument brought another one and then it brought another one and i apologize for that but it was a priceless experience for me well thank you very much you could have gone on all night but now i i'm worried about all your editing well that's your problem not mine i gotta have it exactly exactly thank you so much have a lovely evening thank you so much bye